INCH-POUND

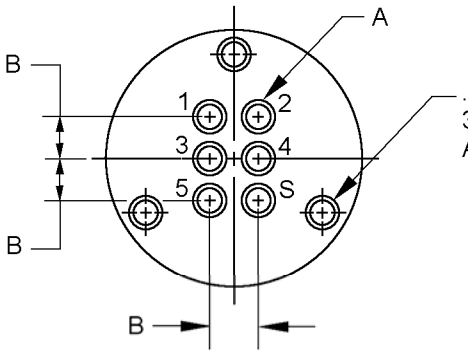
MIL-PRF-27/65D
AMENDMENT 1
12 August 2016
SUPERSEDING
MIL-PRF-27/65D
6 May 2005

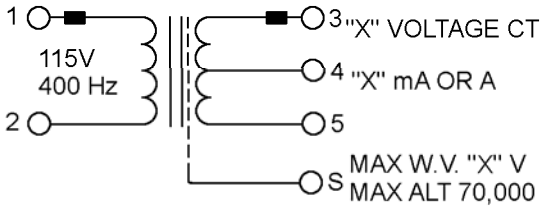
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 100 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

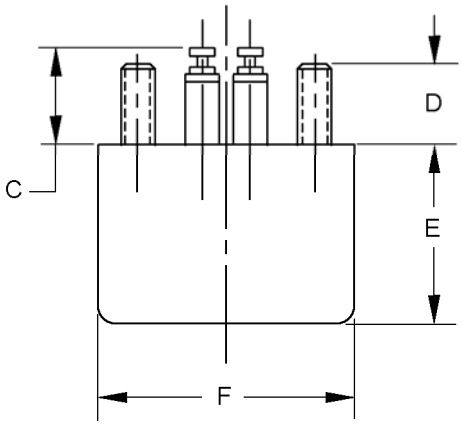
The requirements for acquiring the transformers described herein shall consist of this specification and MIL-PRF-27.





(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING



Dim	Inches	mm
A	.250 DIA. TYP.	6.35 DIA. TYP.
B	.312	7.92
C	.390 MAX	9.91 MAX
D	.344 ±.031	8.74 ±.078
E	2.000 ±.031	50.80 ±.078
F	2.250 ±.031	57.15 ±.078

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown. Ground terminal "S" is black.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 18.7 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: .390 inch (9.91 mm), maximum.

Terminal strength: MIL-STD-202-211, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 75 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 93 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)} \quad x \quad 100$

Shall not exceed 2.5 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 43°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 87°C.

Vibration: MIL-STD-202-204, test condition D.

Marking location: See figure 1.

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VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-PRF-27/48 through MIL-PRF-27/66 inclusive, all parts.

Qualification testing and approval to M27/65-36 shall be sufficient to grant qualification approval to M27/65-01 through M27/65-36.

Part or Identifying Number (PIN) example:

M	27/65	-01
MIL specification item	Specification sheet no.	Dash no.

TABLE I. Electrical ratings.

Dash number	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01	12	8.33	325
02	12.6	7.94	325
03	13	7.69	325
04	15	6.67	325
05	16	6.25	325
06	18	5.56	325
07	20	5.00	325
08	22	4.55	325
09	24	4.17	325
10	26	3.85	325
11	28	3.57	325
12	30	3.33	325
13	33	3.03	325
14	36	2.78	325
15	39	2.56	325
16	43	2.33	325
17	47	2.13	325
18	51	1.96	325
19	56	1.79	325
20	62	1.61	325
21	68	1.47	325
22	75	1.33	325
23	82	1.22	325
24	91	1.10	325
25	100	1.00	325
26	110	0.91	325
27	115	0.87	325
28	120	0.83	325
29	130	0.77	325
30	150	0.67	325
31	180	0.56	325
32	200	0.50	440
33	220	0.45	440
34	240	0.42	440
35	270	0.37	580
36	300	0.33	580

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204, MIL-STD-202-211

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2016-016)

Review activities:
Army – CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

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